

DATA-260 HW3

Rachandeep Kaur

Github Link- <https://github.com/rachandeepkaur/data260-7117.git>

Section 0 — Personal Configuration and Domain

Value	Definition	My value
SID4	Last four digits of SJSU Student ID	7117
PORT_BASE	$8000 + (\text{SID4} \bmod 900)$	$8000 + 817 = 8817$
PREFIX	"s" + SID4	s7117
SEED	SID4	7117
VERIFY_SEED	$260000 + \text{SID4}$	267117
DOMAIN_ID	$\text{SID4} \bmod 8$	5 — Local restaurant inspections

Item	Value
Hardware	MacBook Pro — Intel Core i9-9880H @ 2.30 GHz (8 cores / 16 threads), 32 GB RAM, macOS 26.6.2
Local model	qwen3:8b via Ollama (temperature 0, seed 7117, thinking disabled)
Embedding model	BAAI/bge-small-en-v1.5 (fastembed / ONNXRuntime, 384-dim, 512-token window)
Database	MySQL 8.4.11 (Docker image mysql:8.4, container s7117-mysql), database s7117_rel
Backend / frontend	FastAPI + SQLAlchemy 2 on port 8817; React 18 + React Router 6 (Vite) on port 5173
Tagged commit	hw4

Part 1 - React Client

Login page/component

```

11
12   const handleSubmit = async (event) => {
13     event.preventDefault();
14     setSubmitting(true);
15     setError("");
16     try {
17       await onLogin(email, password);
18     } catch (err) {
19       setError(err.message);
20     } finally {
21       setSubmitting(false);
22     }
23   };
24
25   return (
26     <form className="card form" onSubmit={handleSubmit}>
27       <h2>Inspector Login</h2>
28       {error && <div className="alert">{error}</div>}
29       <label>
30         Email
31         <input type="email" value={email} onChange={(e) => setEmail(e.target.value)} required autoFocus/
32       </label>
33       <label>
34         Password
35         <input type="password" value={password} onChange={(e) => setPassword(e.target.value)} required/
36       </label>
37       <button type="submit" className="btn btn-primary" disabled={submitting}>
38         {submitting ? "Logging in..." : "Login"}
39       </button>
40     </form>
41   );
42 }
43

```

```

const login = async (email, password) => {
  const { data } = await api.login(email, password);
  setUser(data);
  setPage(0);
  navigate("/");
};

const logout = async () => {
  await api.logout().catch(() => {});
  setUser(null);
  navigate("/login");
};

```

Login required

You must be logged in to view and manage inspection records.

[Go to Login](#)

Application												
Filter												
Only show cookies with an issue												
Name	Value	Do...	Path	Exp...	Size	HttpO...	S...	S...	Par...	Cro...	Prio...	
_xsrf	2[d50f3baa]d02437533eda66a...	loc...	/	202...	59						Me...	
ajs_anonymous_id	2f6361e7-9dc9-4b6f-8419-811...	loc...	/	202...	52						Me...	
s7117_session	QoB0SO_beKoifJOBR9XefJnXQ...	loc...	/	202...	56	✓		Lax			Me...	
username-localho...	"2[1:0]10:1788672430]23:usern...	loc...	/	202...	341	✓					Me...	

I. Home Page (Home.jsx) — route /

code > react_client > src > components > Home.jsx > ...

```
1  import { Link } from "react-router-dom";
2
3  export default function Home({ records, total, page, pageSize, onPageChange }) {
4    const lastPage = Math.max(0, Math.ceil(total / pageSize) - 1);
5
6    return (
7      <div className="card">
8        <div className="card-header">
9          <h2>Inspection Records</h2>
10         <span className="muted">{total.toLocaleString()} records in MySQL</span>
11       </div>
12       <table>
13         <thead>
14           <tr>
15             <th>ID</th>
16             <th>Facility Name / ID</th>
17             <th>Program Site Address</th>
18             <th>Actions</th>
19           </tr>
20         </thead>
21         <tbody>
22           {records.map((record) => (
23             <tr key={record.id}>
24               <td>{record.id}</td>
25               <td>{record.facility_name}</td>
26               <td>{record.site_address}</td>
27               <td className="actions">
28                 <Link to={` /update?id=${record.id}`} className="btn">Edit</Link>
29                 <Link to={` /delete?id=${record.id}`} className="btn btn-danger">Delete</Link>
30               </td>
31             </tr>
32           ))}
33           {records.length === 0 && (
34             <tr>
35               <td colSpan={4} className="muted">No inspection records yet.</td>
36             </tr>
37           )}
38         </tbody>
39       </table>
40       <div className="pager">
41         <button className="btn" disabled={page === 0} onClick={() => onPageChange(page - 1)}>Previous<
42         <span>Page {page + 1} of {lastPage + 1}</span>
43         <button className="btn" disabled={page >= lastPage} onClick={() => onPageChange(page + 1)}>Next<
44       </div>
45     </div>
46   );
```

```
const loadRecords = useCallback(async () => {
  try {
    const { data, headers } = await api.listRecords(PAGE_SIZE, page * PAGE_SIZE);
    setRecords(data);
    setTotal(Number(headers.get("X-Total-Count") ?? data.length));
    setError("");
  } catch (err) {
    handleApiError(err);
  }
}, [page, handleApiError]);
```

restaurant inspections

[Home](#)
[Add Inspection Record](#)
Signed in as Rachandeep Kaur
[Logout](#)

Inspection Records

5,000 records in MySQL

ID	Facility Name / ID	Program Site Address	Actions	
5000	FA0470962 - LA CANTINAS	8859 WINCHESTER BLVD, SAN JOSE, CA 95127	Edit	Delete
4999	FA0422276 - RED BURGERS	4407 ALMADEN EXPY, SAN JOSE, CA 95128	Edit	Delete
4998	FA0536430 - LA PIZZERIA	3057 MCKEE RD, SAN JOSE, CA 95120	Edit	Delete
4997	FA0599994 - LA CURRY HOUSE	6119 MONTEREY RD, SAN JOSE, CA 95123	Edit	Delete
4996	FA0179358 - VALLEY WOK	8078 SENTER RD, SAN JOSE, CA 95117	Edit	Delete
4995	FA0907630 - MISSION DINER	6780 WINCHESTER BLVD, SAN JOSE, CA 95116	Edit	Delete
4994	FA0272023 - BAY RAMEN	951 E SANTA CLARA ST, SAN JOSE, CA 95136	Edit	Delete
4993	FA0162094 - GREEN RAMEN	2378 SARATOGA AVE, SAN JOSE, CA 95128	Edit	Delete
4992	FA0177684 - LITTLE TACOS	8362 KING RD, SAN JOSE, CA 95124	Edit	Delete
4991	FA0199437 - GREEN DELI	7647 BLOSSOM HILL RD, SAN JOSE, CA 95128	Edit	Delete
4990	FA0155324 - PHO CURRY HOUSE	9896 STEVENS CREEK BLVD, SAN JOSE, CA 95131	Edit	Delete
4989	FA0613556 - EL GRILL	1701 MCKEE RD, SAN JOSE, CA 95127	Edit	Delete
4988	FA0585706 - ROYAL SUSHI	7157 CAPITOL EXPY, SAN JOSE, CA 95122	Edit	Delete
4987	FA0619102 - LUCKY CURRY HOUSE	9890 E SANTA CLARA ST, SAN JOSE, CA 95133	Edit	Delete
4986	FA0861213 - GREEN WOK	329 KING RD, SAN JOSE, CA 95126	Edit	Delete

II. Add a New Record (CreateRecord.jsx) — route /create

code > react_client > src > components > CreateRecord.jsx > ...

```
1  import { useState } from "react";
2  import { useNavigate } from "react-router-dom";
3
4  export default function CreateRecord({ onAddRecord }) {
5    const navigate = useNavigate();
6    const [facilityName, setFacilityName] = useState("");
7    const [siteAddress, setSiteAddress] = useState("");
8    const [error, setError] = useState("");
9
10   const handleSubmit = async (event) => {
11     event.preventDefault();
12     try {
13       // The id is assigned by MySQL AUTO_INCREMENT on the backend.
14       await onAddRecord({ facility_name: facilityName, site_address: siteAddress });
15       navigate("/");
16     } catch (err) {
17       setError(err.message);
18     }
19   };
20
21   return (
22     <form className="card form" onSubmit={handleSubmit}>
23       <h2>Add Inspection Record</h2>
24       {error && <div className="alert">{error}</div>}
25       <label>
26         Facility Name / ID
27         <input
28           value={facilityName}
29           onChange={(e) => setFacilityName(e.target.value)}
30           placeholder="e.g., FA0206933 - YUMMY KITCHEN"
31           required
32           autoFocus
33         />
34       </label>
35       <label>
36         Program Site Address
37         <input
38           value={siteAddress}
39           onChange={(e) => setSiteAddress(e.target.value)}
40           placeholder="e.g., 1711 BRANHAM LN A9, SAN JOSE, CA 95118"
41           required
42         />
43       </label>
44       <button type="submit" className="btn btn-primary">Add Inspection Record</button>
45     </form>
46   );
47 }
```

```
0  const addRecord = async (record) => {
1    await api.createRecord(record);
2    setPage(0); // newest first, so the new record is on page 1
3    await loadRecords();
4  };
5
```

Add Inspection Record

Facility Name / ID

e.g., FA0206933 - YUMMY KITCHEN

Program Site Address

e.g., 1711 BRANHAM LN A9, SAN JOSE, CA 95118

Add Inspection Record

III. Update Record (UpdateRecord.jsx) — route /update

code > react_client > src > components > UpdateRecord.jsx > UpdateRecord

```
5  export default function UpdateRecord({ onUpdateRecord, fetchRecord }) {
6    const navigate = useNavigate();
7    const [searchParams, setSearchParams] = useSearchParams();
8    const recordId = searchParams.get("id");
9    const [idInput, setIdInput] = useState("");
10   const [facilityName, setFacilityName] = useState("");
11   const [siteAddress, setSiteAddress] = useState("");
12   const [loaded, setLoaded] = useState(false);
13   const [error, setError] = useState("");
14
15   // Prefill the form with the record's current values from MySQL. `ignore`
16   // drops a stale response (e.g. StrictMode's second run) so it can't
17   // overwrite what the user has already typed.
18   useEffect(() => {
19     if (!recordId) return undefined;
20     let ignore = false;
21     setLoaded(false);
22     fetchRecord(recordId)
23       .then((record) => {
24         if (ignore) return;
25         setFacilityName(record.facility_name);
26         setSiteAddress(record.site_address);
27         setLoaded(true);
28         setError("");
29       })
30       .catch((err) => { if (!ignore) setError(err.message); });
31     return () => { ignore = true; };
32   }, [recordId, fetchRecord]);
33
34   const handleSubmit = async (event) => {
35     event.preventDefault();
36     try {
37       await onUpdateRecord(recordId, { facility_name: facilityName, site_address: siteAddress });
38       navigate("/");
39     } catch (err) {
40       setError(err.message);
41     }
42   };
43
44   if (!recordId) {
45     return (
46       <form className="card form" onSubmit={(e) => { e.preventDefault(); setSearchParams({ id: idInput }); }}>
47         <h2>Update Inspection Record</h2>
48         <label>
49           Record ID
50           <input type="number" min="1" value={idInput} onChange={(e) => setIdInput(e.target.value)} required />
51         </label>
52         <button type="submit" className="btn">Load Record</button>
53       </form>
54     );
55   }
56 }
```

Update Inspection Record #5000

Facility Name / ID

FA0470962 - LA CANTINAS

Program Site Address

8859 WINCHESTER BLVD, SAN JOSE, CA 95127

Update Inspection Record

IV. Delete the Record (DeleteRecord.jsx) — route /delete

```

export default function DeleteRecord({ onDeleteRecord, fetchRecord }) {
  const navigate = useNavigate();
  const [searchParams, setSearchParams] = useSearchParams();
  const recordId = searchParams.get("id");
  const [idInput, setIdInput] = useState("");
  const [record, setRecord] = useState(null);
  const [error, setError] = useState("");

  useEffect(() => {
    if (!recordId) return undefined;
    let ignore = false;
    setRecord(null);
    fetchRecord(recordId)
      .then((data) => { if (!ignore) { setRecord(data); setError(""); } })
      .catch((err) => { if (!ignore) setError(err.message); });
    return () => { ignore = true; };
  }, [recordId, fetchRecord]);

  const handleDelete = async () => {
    try {
      await onDeleteRecord(recordId);
      navigate("/");
    } catch (err) {
      setError(err.message);
    }
  };

  if (!recordId) {
    return (
      <form className="card form" onSubmit={(e) => { e.preventDefault(); setSearchParams({ id: idInput }); }}>
        <h2>Delete Inspection Record</h2>
        <label>
          Record ID
          <input type="number" min="1" value={idInput} onChange={(e) => setIdInput(e.target.value)} required />
        </label>
        <button type="submit" className="btn">Load Record</button>
      </form>
    );
  }
}

```

```

return (
  <div className="card form">
    <h2>Delete Inspection Record #{recordId}</h2>
    {error && <div className="alert">{error}</div>}
    {record && (
      <dl>
        <dt>Facility Name / ID</dt>
        <dd>{record.facility_name}</dd>
        <dt>Program Site Address</dt>
        <dd>{record.site_address}</dd>
      </dl>
    )}
    <div className="actions">
      <button className="btn btn-danger" onClick={handleDelete} disabled={!record}>Delete Inspection Record</button>
      <button className="btn" onClick={() => navigate("/")}>Cancel</button>
    </div>
  </div>
);
}

```

Delete Inspection Record #5000

Facility Name / ID
FA0470962 - LA CANTINAS

Program Site Address
8859 WINCHESTER BLVD, SAN JOSE, CA 95127

Delete Inspection Record

Cancel

V. Props, hooks and React Router

Code — App.jsx — React Router routes with props

```
{error && user && <div className="alert">{error}</div>}
<Routes>
  <Route
    path="/"
    element={guard(
      <Home records={records} total={total} page={page} pageSize={PAGE_SIZE} onPageChange={setPage} />
    )}
  />
  <Route path="/login" element={<Login onLogin={login} user={user} />} />
  <Route path="/create" element={guard(<CreateRecord onAddRecord={addRecord} />)} />
  <Route
    path="/update"
    element={guard(<UpdateRecord onUpdateRecord={updateRecord} fetchRecord={fetchRecord} />)}
  />
  <Route
    path="/delete"
    element={guard(<DeleteRecord onDeleteRecord={deleteRecord} fetchRecord={fetchRecord} />)}
  />
</Routes>
</main>
```

```
code > react_client > src > main.jsx
1  import React from "react";
2  import ReactDOM from "react-dom/client";
3  import { BrowserRouter } from "react-router-dom";
4  import App from "../App.jsx";
5  import "../styles.css";
6
7  ReactDOM.createRoot(document.getElementById("root")).render(
8    <React.StrictMode>
9      <BrowserRouter>
10        <App />
11      </BrowserRouter>
12    </React.StrictMode>
13  );
14
```

Part 2 — MySQL Persistence and Server-Side Sessions

Database s7117_rel and tables

```
7  -- Primary domain entity (DOMAIN_ID 5 - local restaurant inspections).
8  CREATE TABLE IF NOT EXISTS inspections (
9    id          INT          NOT NULL AUTO_INCREMENT,
10   facility_name VARCHAR(255) NOT NULL, -- primary field
11   site_address VARCHAR(255) NOT NULL, -- secondary field
12   PRIMARY KEY (id)
13 ) ENGINE = InnoDB;
```

```
-- Authentication / server-side sessions.
CREATE TABLE IF NOT EXISTS users (
  id          INT          NOT NULL AUTO_INCREMENT,
  name        VARCHAR(120) NOT NULL,
  email       VARCHAR(255) NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  PRIMARY KEY (id),
  UNIQUE KEY uq_users_email (email)
) ENGINE = InnoDB;
```

```

● (.venv) (base) pc@pcs-MacBook-Pro data260-7117 % docker start s7117-mysql
s7117-mysql
● (.venv) (base) pc@pcs-MacBook-Pro data260-7117 % make setup-db
.venv/bin/python code/hw04/setup_db.py
Connected to MySQL 8.4.11
s7117_rel tables: ['inspection_violations', 'inspections', 'sessions', 'users']
○ (.venv) (base) pc@pcs-MacBook-Pro data260-7117 % █

```

SQLAlchemy connection — db_session_based

Users, login and server-side sessions (HTTP-only cookie)

```

64
65 class User(Base):
66     __tablename__ = "users"
67
68     id: Mapped[int] = mapped_column(Integer, primary_key=True, autoincrement=True)
69     name: Mapped[str] = mapped_column(String(120), nullable=False)
70     email: Mapped[str] = mapped_column(String(255), nullable=False, unique=True)
71     password_hash: Mapped[str] = mapped_column(String(255), nullable=False)
72
73
74 class UserSession(Base):
75     __tablename__ = "sessions"
76
77     # The opaque session token itself; the browser cookie holds only this value.
78     id: Mapped[str] = mapped_column(String(64), primary_key=True)
79     user_id: Mapped[int] = mapped_column(ForeignKey("users.id", ondelete="CASCADE"), nullable=False)
80     created_at: Mapped[datetime] = mapped_column(DateTime, nullable=False)
81     expires_at: Mapped[datetime] = mapped_column(DateTime, nullable=False)
82
83     user: Mapped[User] = relationship()
84

```

```

SESSION_COOKIE_NAME = "s7117_session"
SESSION_TTL = timedelta(hours=int(os.getenv("SESSION_TTL_HOURS", "8")))
# Local dev runs over plain http; set COOKIE_SECURE=1 when served over https.
COOKIE_SECURE = os.getenv("COOKIE_SECURE", "0") == "1"

_PBKDF2_ITERATIONS = 200_000

def hash_password(password: str) -> str:
    salt = secrets.token_hex(16)
    digest = hashlib.pbkdf2_hmac("sha256", password.encode(), bytes.fromhex(salt), _PBKDF2_ITERATIONS)
    return f"pbkdf2_sha256${_PBKDF2_ITERATIONS}${salt}${digest.hex()}"

def verify_password(password: str, stored_hash: str) -> bool:
    try:
        _algo, iterations, salt, expected = stored_hash.split("$")
    except ValueError:
        return False
    digest = hashlib.pbkdf2_hmac("sha256", password.encode(), bytes.fromhex(salt), int(iterations))
    return hmac.compare_digest(digest.hex(), expected)

def utcnow() -> datetime:
    # MySQL DATETIME has no timezone; store naive UTC consistently.
    return datetime.now(timezone.utc).replace(tzinfo=None)

def create_session(db_session_basede26: Session, user: User) -> UserSession:
    now = utcnow()
    user_session = UserSession(
        id=secrets.token_urlsafe(32),
        user_id=user.id,
        created_at=now,
        expires_at=now + SESSION_TTL,
    )
    db_session_basede26.add(user_session)
    db_session_basede26.commit()
    return user_session

def get_current_user(
    s7117_session: str | None = Cookie(default=None),
    db_session_basede26: Session = Depends(get_db),
    \ ~ User:

```

```

@router.post("/login", response_model=UserOut)
def login(payload: LoginIn, response: Response, db_session_basede26: Session = Depends(get_db)) -> User:
    user = db_session_basede26.execute(
        select(User).where(User.email == payload.email.lower())
    ).scalar_one_or_none()
    if user is None or not verify_password(payload.password, user.password_hash):
        raise HTTPException(status_code=401, detail="Invalid email or password")

    user_session = create_session(db_session_basede26, user)
    response.set_cookie(
        key=SESSION_COOKIE_NAME,
        value=user_session.id,
        httponly=True,
        samesite="lax",
        secure=COOKIE_SECURE,
        max_age=int(SESSION_TTL.total_seconds()),
        path="/",
    )
    return user

```

http://127.0.0.1:8817/api/auth/login

POST http://127.0.0.1:8817/api/auth/login

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Headers Hide auto-generated headers

Key	Value
Cookie	s7117_session=NvM64aI_Pc8n3cuMQEC1HUfCaTpeUBvuupueUJyo-jk
Postman-Token	<calculated when request is sent>
Content-Type	application/json
Content-Length	<calculated when request is sent>
Host	<calculated when request is sent>
User-Agent	PostmanRuntime/7.56.1
Accept	*/*
Accept-Encoding	gzip, deflate, br
Connection	keep-alive

Body Cookies (1) Headers (5) Test Results

Status: 200 OK Time: 177 ms Size: 301 B

Pretty Raw Preview Visualize JSON

```

1  {
2    "id": 1,
3    "name": "Rachandeep Kaur",
4    "email": "demo@s7117.dev"
5  }

```

```
mysql> SELECT * FROM sessions;
```

id	user_id	created_at	expires_at
3sDXgPrAlgQdZsLxa2XkZ5FPcfiUI4Q-gjaiZoxXVVQ	1	2026-09-28 09:01:50	2026-09-28 17:01:50
bCuWzXCDXtrQfnSSlxCtuuA1hJ7-dG14RbXxL3zu8WE	1	2026-09-28 08:09:39	2026-09-28 16:09:39
ELsTLjvpyKnkwy6LnT62_VV0biZyiHnpbYCSprH9CzI	1	2026-09-28 08:08:45	2026-09-28 16:08:45
kdLIlgAman33cTMVnnIdho8no_Y8500b7hvpavmVNFY	1	2026-09-28 08:20:14	2026-09-28 16:20:14
M3B181YnDbbruLTVkFK61_zruAUfAbZZ9qEXtRdJvJUM	1	2026-09-28 09:01:35	2026-09-28 17:01:35
Msu3X0bCB1_g0vu9JJw4EiB95yZUpxe6iDhvggE4Agk	1	2026-09-28 08:08:03	2026-09-28 16:08:03
NvM64ai_Pc8n3cuMQEC1HUfCaTpeUBvuupueUJyo-jk	1	2026-09-28 19:26:08	2026-09-29 03:26:08
Q726Jt72N9UGTTfQi5IOMRZkJ4FGGurQ0CpV3LZdrJ0	1	2026-09-28 09:04:58	2026-09-28 17:04:58
QoB0S0_beKoifJOBR9XefJnXQ7iXDld1hifKLpMApIQ	1	2026-09-28 18:50:15	2026-09-29 02:50:15
wD1aYPNq6Ne5M0W1f62yEurMPN0cd6CPgRtuELhD5A	1	2026-09-28 08:20:55	2026-09-28 16:20:55
ygapLGx0gJ5TJRTgceCYhHdpHI9P9kZaQY_d1HI4U9Q	1	2026-09-28 08:06:36	2026-09-28 16:06:36

```
11 rows in set (0.00 sec)
```

After deleting cookies in the postman

http://127.0.0.1:8817/api/records

GET http://127.0.0.1:8817/api/records

Params Authorization Headers (6) Body Pre-request Script Tests Settings

Headers Hide auto-generated headers

Key	Value
☒ Postman-Token	<calculated when request is sent>
☒ Host	<calculated when request is sent>
☒ User-Agent	PostmanRuntime/7.56.1
☒ Accept	*/*
☒ Accept-Encoding	gzip, deflate, br
☒ Connection	keep-alive
Key	Value

Body Cookies Headers (4) Test Results

Pretty Raw Preview Visualize JSON

```
1 {
2   "detail": "Login required"
3 }
```

Status: 401 Unauthorized Time: 14 ms Size:

CRUD API for inspection records

Add a new record (POST /api/records)

```
@router.post("", response_model=RecordOut, status_code=201)
def create_record(
    payload: RecordIn,
    _user: User = Depends(get_current_user),
    db_session_base26: Session = Depends(get_db),
) -> Inspection:
    record = Inspection(facility_name=payload.facility_name.strip(), site_address=payload.site_address.strip())
    db_session_base26.add(record)
    db_session_base26.commit() # MySQL AUTO_INCREMENT assigns record.id
    return record
```

HTTP <http://127.0.0.1:8817/api/records>

POST <http://127.0.0.1:8817/api/records>

Params Authorization **Headers (9)** Body ● Pre-request Script Tests Settings

Headers Hide auto-generated headers

Key	Value
☒ Cookie	s7117_session=rlyMAz8a5EXyGll3onXC9Rdt4m-eLJgdXN5-9W8FBpQ
☒ Postman-Token	<calculated when request is sent>
☒ Content-Type	application/json
☒ Content-Length	<calculated when request is sent>
☒ Host	<calculated when request is sent>
☒ User-Agent	PostmanRuntime/7.56.1
☒ Accept	/*/*
☒ Accept-Encoding	gzip, deflate, br
☒ Connection	keep-alive

Body Cookies (1) Headers (4) Test Results Status: 201 Created Time: 19 ms

Pretty Raw Preview Visualize JSON

```
1
2  "facility_name": "FA0206933 - YUMMY KITCHEN",
3  "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118",
4  "id": 5004
5
```

View all records (GET /api/records)

```
@router.get("", response_model=list[RecordOut])
def list_records(
    response: Response,
    limit: Optional[int] = Query(default=None, ge=1, le=10000),
    offset: int = Query(default=0, ge=0),
    _user: User = Depends(get_current_user),
    db_session_based26: Session = Depends(get_db),
) -> list[Inspection]:
    """Newest first, so a just-created record shows at the top of the list."""
    total = db_session_based26.scalar(select(func.count()).select_from(Inspection))
    response.headers["X-Total-Count"] = str(total)
    stmt = select(Inspection).order_by(Inspection.id.desc()).offset(offset)
    if limit is not None:
        stmt = stmt.limit(limit)
    return list[Inspection](db_session_based26.execute(stmt).scalars())
```

HTTP http://127.0.0.1:8817/api/records

GET http://127.0.0.1:8817/api/records Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

Headers Hide auto-generated headers

Key	Value	Bulk Edit
☒ Cookie	s7117_session=rtyMAz8a5EXyGll3onXC9Rdt4m-eLJgdXN5-9W8FBpQ	
☒ Postman-Token	<calculated when request is sent>	
☒ Content-Type	application/json	
☒ Content-Length	<calculated when request is sent>	
☒ Host	<calculated when request is sent>	
☒ User-Agent	PostmanRuntime/7.56.1	
☒ Accept	*/*	
☒ Accept-Encoding	gzip, deflate, br	
☒ Connection	keep-alive	

body Cookies (1) Headers (5) Test Results Status: 200 OK Time: 455 ms Size: 532.05 KB Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   {
3     "facility_name": "FA0206933 - YUMMY KITCHEN",
4     "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118",
5     "id": 5004
6   },
7   {
8     "facility_name": "FA0470962 - LA CANTINAS",
9     "site_address": "8859 WINCHESTER BLVD, SAN JOSE, CA 95127",
10    "id": 5000
11  },
12  {
13    "facility_name": "FA0422276 - RED BURGERS",
14    "site_address": "4407 ALMADEN EXPY, SAN JOSE, CA 95128",
15    "id": 4999
16  },
17  {
18    "facility_name": "FA0536430 - LA PIZZERIA",
19    "site_address": "3057 MCKEE RD, SAN JOSE, CA 95120",
20    "id": 4998
21  }
22 }
```

View a record by ID (GET /api/records/{id})

```
@router.get("/{record_id}", response_model=RecordOut)
def get_record(
    record_id: int,
    _user: User = Depends(get_current_user),
    db_session_basede26: Session = Depends(get_db),
) -> Inspection:
    return _get_or_404(db_session_basede26, record_id)
```

http://127.0.0.1:8817/api/records/2000

GET http://127.0.0.1:8817/api/records/2000 Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

Hide auto-generated headers

Key	Value	Bulk Edit
☒ Cookie	s7117_session=rlyMAz8a5EXyGll3onXC9Rdt4m-eLjgdXN5-9W8FBpQ	
☒ Postman-Token	<calculated when request is sent>	
☒ Content-Type	application/json	
☒ Content-Length	<calculated when request is sent>	
☒ Host	<calculated when request is sent>	
☒ User-Agent	PostmanRuntime/7.56.1	
☒ Accept	*/*	
☒ Accept-Encoding	gzip, deflate, br	
☒ Connection	keep-alive	

ody Cookies (1) Headers (4) Test Results

Status: 200 OK Time: 12 ms Size: 240 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "facility_name": "FA0215571 - LUCKY RAMEN",
3   "site_address": "8499 STEVENS CREEK BLVD, SAN JOSE, CA 95127",
4   "id": 2000
5 }
```

Update record details (PUT /api/records/{id})

```

01
02
03 @router.put("/{record_id}", response_model=RecordOut)
04 def update_record(
05     record_id: int,
06     payload: RecordUpdate,
07     _user: User = Depends(get_current_user),
08     db_session_basede26: Session = Depends(get_db),
09 ) -> Inspection:
10     record = _get_or_404(db_session_basede26, record_id)
11     record.facility_name = payload.facility_name.strip()
12     if payload.site_address is not None:
13         record.site_address = payload.site_address.strip()
14     db_session_basede26.commit()
15     return record
16
17

```

http://127.0.0.1:8817/api/records/2000

PUT http://127.0.0.1:8817/api/records/2000

Params Authorization Headers (9) Body Pre-request Script Tests Settings

none form-data x-www-form-urlencoded raw binary JSON

Save Beautify

```

1 {
2   "facility_name": "FA0215571 - LUCKY RAMEN NAGI",
3   "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118"
4 }
5

```

Body Cookies (1) Headers (4) Test Results

Status: 200 OK Time: 21 ms Size: 240 B Save Response

Pretty Raw Preview Visualize JSON

```

1 {
2   "facility_name": "FA0215571 - LUCKY RAMEN NAGI",
3   "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118",
4   "id": 2000
5 }

```

Delete a record (DELETE /api/records/{id})

```
7
8 @router.delete("/{record_id}", response_model=RecordOut)
9 def delete_record(
10     record_id: int,
11     _user: User = Depends(get_current_user),
12     db_session_basede26: Session = Depends(get_db),
13 ) -> Inspection:
14     record = _get_or_404(db_session_basede26, record_id)
15     deleted = RecordOut(id=record.id, facility_name=record.facility_name, site_address=record.site_address)
16     db_session_basede26.delete(record) # cascades to its inspection_violations
17     db_session_basede26.commit()
18     return deleted
19
```

http://127.0.0.1:8817/api/records/5004 Save

DELETE Send

Params Authorization **Headers (9)** Body Pre-request Script Tests Settings Cookies

Hide auto-generated headers

Key	Value	Bulk Edit
☒ Cookie ⓘ	s7117_session=rlyMAz8a5EXyGli3onXC9Rdt4m-eLJgdXN5-9W8FBpQ	
☒ Postman-Token ⓘ	<calculated when request is sent>	
☒ Content-Type ⓘ	application/json	
☒ Content-Length ⓘ	<calculated when request is sent>	
☒ Host ⓘ	<calculated when request is sent>	
☒ User-Agent ⓘ	PostmanRuntime/7.56.1	
☒ Accept ⓘ	*/*	
☒ Accept-Encoding ⓘ	gzip, deflate, br	
☒ Connection ⓘ	keep-alive	

Body **Cookies (1)** **Headers (4)** Test Results Status: 200 OK Time: 21 ms Size: 237 B Save Response

Pretty Raw Preview Visualize JSON ⌵ ⌵

```
1 {
2   "facility_name": "FA0206933 - YUMMY KITCHEN",
3   "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118",
4   "id": 5004
5 }
```

Part 3 — N+1 Measurement and Query Tuning

3.1 Seed 5,000 primary rows and 200 related rows (SEED = 7117)

```

def generate(rng: random.Random) -> tuple[list[dict], list[dict]]:
    inspections = []
    for i in range(1, N_INSPECTIONS + 1):
        fa_number = f"FA{rng.randint(100000, 999999):07d}"
        name = f"{rng.choice(NAME_WORDS_A)} {rng.choice(NAME_WORDS_B)}"
        address = f"{rng.randint(10, 9999)} {rng.choice(STREETS)}, SAN JOSE, CA {rng.choice(ZIPS)}"
        inspections.append({"id": i, "facility_name": f"{fa_number} - {name}", "site_address": address})

    start = date(2025, 1, 1)
    violations = []
    for j in range(1, N_VIOLATIONS + 1):
        code, description, severity, points = rng.choice(VIOLATION_TYPES)
        violations.append({
            "id": j,
            "inspection_id": rng.randint(1, VIOLATION_TARGET_MAX_ID),
            "violation_code": code,
            "description": description,
            "severity": severity,
            "points_deducted": points,
            "observed_on": start + timedelta(days=rng.randint(0, 600)),
        })
    return inspections, violations

```

```

CREATE TABLE IF NOT EXISTS inspection_violations (
    id INT NOT NULL AUTO_INCREMENT,
    inspection_id INT NOT NULL,
    violation_code VARCHAR(16) NOT NULL,
    description VARCHAR(255) NOT NULL,
    severity VARCHAR(16) NOT NULL,
    points_deducted INT NOT NULL,
    observed_on DATE NOT NULL,
    PRIMARY KEY (id)
) ENGINE = InnoDB;

```

```

4 ===== [2026-09-28T08:20:08Z] $ .venv/bin/python code/hw04/setup_db.py
5 Connected to MySQL 8.4.11
6 s7117_rel tables: ['inspection_violations', 'inspections', 'sessions', 'users']
7 ----- exit=0 [2026-09-28T08:20:09Z]
8
9 ===== [2026-09-28T08:20:09Z] $ .venv/bin/python code/hw04/seed_data.py
10 {
11     "seed": 7117,
12     "inspections": 5000,
13     "violations": 200,
14     "inspections_with_violations": 136,
15     "violations_in_first_page": {
16         "10": 9,
17         "50": 45,
18         "200": 162
19     },
20     "generated_data_sha256": "392e4dd10634ccee2b464ec0c5115d098adbc0cbfcdcae796b43ff7fc3a6944",
21     "sample_inspections": [
22         {
23             "id": 1,
24             "facility_name": "FA0183034 - GOLDEN DINER",
25             "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112"
26         },
27         {
28             "id": 2,
29             "facility_name": "FA0958065 - VALLEY BURGERS",
30             "site_address": "455 SENTER RD, SAN JOSE, CA 95124"
31         },
32         {
33             "id": 3,
34             "facility_name": "FA0956037 - ROYAL RAMEN",
35             "site_address": "2085 CAPITOL EXPY, SAN JOSE, CA 95127"
36         }
37     ],
38     "sample_violations": [
39         {
40             "id": 1,
41             "inspection_id": 9,
42             "violation_code": "9",
43             "description": "Improper cooling methods",
44             "severity": "MAJOR",
45             "points_deducted": 4,
46             "observed_on": "2026-04-17"
47         },
48         {
49             "id": 2,
50             "inspection_id": 134,
51             "violation_code": "7",
52             "description": "Improper hot holding temperature (below 135F)",
53             "severity": "MAJOR",
54             "points_deducted": 4,

```

3.2 Related rows associated with the primary entity and included in the list response

```
def _serialize(inspection: Inspection, violations: list[Violation]) -> dict:
    return {
        "id": inspection.id,
        "facility_name": inspection.facility_name,
        "site_address": inspection.site_address,
        "violations": [
            {
                "id": v.id,
                "violation_code": v.violation_code,
                "description": v.description,
                "severity": v.severity,
                "points_deducted": v.points_deducted,
                "observed_on": v.observed_on.isoformat(),
            }
            for v in violations
        ],
    }
```

http://127.0.0.1:8817/api/nplus1/naive?fixed&page_size=10

GET http://127.0.0.1:8817/api/nplus1/naive?fixed&page_size=10

Params Authorization Headers (9) Body Pre-request Script Tests Settings

Hide auto-generated headers

Key	Value	Bulk Edit
☒ Cookie	s7117_session=rlyMAz8a5EXyGll3onXC9Rdt4m-eLJgdXN5-9W8FBpQ	
☒ Postman-Token	<calculated when request is sent>	
☒ Content-Type	application/json	
☒ Content-Length	<calculated when request is sent>	
☒ Host	<calculated when request is sent>	
☒ User-Agent	PostmanRuntime/7.56.1	
☒ Accept	*/*	
☒ Accept-Encoding	gzip, deflate, br	
☒ Connection	keep-alive	

Body Cookies (1) Headers (6) Test Results

Status: 200 OK Time: 31 ms Size: 2.8 KB Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "version": "fixed",
3   "page_size": 10,
4   "offset": 0,
5   "sql_queries": 2,
6   "record_count": 10,
7   "violation_count": 9,
8   "items": [
9     {
10      "id": 1,
11      "facility_name": "FA0183034 - GOLDEN DINER",
12      "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13      "violations": []
14    },
15    {
16      "id": 2,
17      "facility_name": "FA0958065 - VALLEY BURGERS",
18      "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19      "violations": [
20        {
21          "id": 180
```

3.3 Naive list endpoint (one extra query per record)

GET /api/nplus1/naive

```

@router.get("/naive", response_model=PageOut)
def list_naive(
    response: Response,
    page_size: int = Query(default=10, ge=1, le=500),
    offset: int = Query(default=0, ge=0),
    _user: User = Depends(get_current_user),
    db_session_base26: Session = Depends(get_db),
) -> dict:
    """Intentionally naive: one extra SELECT per inspection on the page."""
    started = time.perf_counter()
    with count_queries() as counter:
        inspections = db_session_base26.execute(
            select(Inspection).order_by(Inspection.id).limit(page_size).offset(offset)
        ).scalars().all()
        items = []
        for inspection in inspections:
            violations = db_session_base26.execute(
                select(Violation)
                .where(Violation.inspection_id == inspection.id)
                .order_by(Violation.id)
            ).scalars().all()
            items.append(_serialize(inspection, violations))
    return _page_response(response, "naive", page_size, offset, counter.count, items, started)

```

```

def install(engine: Engine) -> None:
    @event.listens_for(engine, "before_cursor_execute")
    def _count(conn, cursor, statement, parameters, context, executemany)
        counter = getattr(_local, "counter", None)
        if counter is not None:
            counter.count += 1
            counter.statements.append(statement)

```

http://127.0.0.1:8817/api/nplus1/naive?page_size=10

GET http://127.0.0.1:8817/api/nplus1/naive?page_size=10 Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

Headers Hide auto-generated headers

Key	Value	Bulk Edit
☒ Cookie	s7117_session=rlyMAz8a5EXyGll3onXC9Rdt4m-eLJgdXN5-9W8FBpQ	
☒ Postman-Token	<calculated when request is sent>	
☒ Content-Type	application/json	
☒ Content-Length	<calculated when request is sent>	

Body Cookies (1) Headers (6) Test Results Status: 200 OK Time: 40 ms Size: 2.8 KB Save Response

Pretty Raw Preview Visualize JSON

```
1  {
2    "version": "naive",
3    "page_size": 10,
4    "offset": 0,
5    "sql_queries": 11,
6    "record_count": 10,
7    "violation_count": 9,
8    "items": [
9      {
10       "id": 1,
11       "facility_name": "FA0183034 - GOLDEN DINER",
12       "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13       "violations": []
14     },
15     {
16       "id": 2,
17       "facility_name": "FA0958065 - VALLEY BURGERS",
18       "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19       "violations": [
20         {
21           "id": 180,
22           "violation_code": "23",
23           "description": "Vermin or animal contamination observed",
24           "severity": "MAJOR",
25           "points_deducted": 4,
26           "observed_on": "2025-09-24"
27         }
28       ]
29     }
30   ]
}
```

3.4 Measure the naive endpoint at page sizes 10, 50, 200 (30 requests each)

```

rows: list[dict] = []
for page_size in PAGE_SIZES:
    for _ in range(WARMUP):
        for version in VERSIONS:
            one_request(session, version, page_size)
    for i in range(1, N_REQUESTS + 1):
        for version in VERSIONS:
            resp, elapsed_ms = one_request(session, version, page_size)
            body = resp.json()
            rows.append({
                "timestamp_utc": datetime.now(timezone.utc).isoformat(),
                "page_size": page_size,
                "version": version,
                "request_index": i,
                "status": resp.status_code,
                "sql_queries": int(resp.headers["X-SQL-Query-Count"]),
                "client_latency_ms": round(elapsed_ms, 3),
                "server_time_ms": float(resp.headers["X-Server-Time-Ms"]),
                "records_returned": body["record_count"],
                "violations_returned": body["violation_count"],
                "items_sha256": hashlib.sha256(
                    json.dumps(body["items"], sort_keys=True).encode()
                ).hexdigest()[:16],
            })
print(f"[{datetime.now(timezone.utc).isoformat()}] page_size={page_size}: "
      f"{N_REQUESTS} naive + {N_REQUESTS} fixed requests done")

```

Page size	Version	SQL stmts/req	p50 (ms)	p95 (ms)	p99 (ms)
10	naive	11	40.98	83.14	104.87
50	naive	51	109.7	134.98	145.61
200	naive	201	439.71	582.49	653.19

3.5 Fix the N+1 issue with eager loading and re-measure

```

@router.get("/fixed", response_model=PageOut)
def list_fixed(
    response: Response,
    page_size: int = Query(default=10, ge=1, le=500),
    offset: int = Query(default=0, ge=0),
    _user: User = Depends(get_current_user),
    db_session_based26: Session = Depends(get_db),
) -> dict:
    """Fixed: eager-load all violations for the page in one IN (...) query."""
    started = time.perf_counter()
    with count_queries() as counter:
        inspections = db_session_based26.execute(
            select(Inspection)
            .options(selectinload(Inspection.violations))
            .order_by(Inspection.id)
            .limit(page_size)
            .offset(offset)
        ).scalars().all()
        items = [_serialize(inspection, inspection.violations) for inspection in inspections]
    return _page_response(response, "fixed", page_size, offset, counter.count, items, started)

```

Page size	Version	SQL stmts/req	p50 (ms)	p95 (ms)	p99 (ms)
10	fixed	2	23.46	46.82	51.83
50	fixed	2	17.17	29.62	34.64
200	fixed	2	30.78	52.84	61.17

Postman interface showing a GET request to `http://127.0.0.1:8817/api/nplus1/fixed?page_size=10`. The response is a JSON array of two facility objects, each containing a list of violations.

```

1  {
2    "version": "fixed",
3    "page_size": 10,
4    "offset": 0,
5    "sql_queries": 2,
6    "record_count": 10,
7    "violation_count": 9,
8    "items": [
9      {
10       "id": 1,
11       "facility_name": "FA0183934 - GOLDEN DINER",
12       "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13       "violations": []
14     },
15     {
16       "id": 2,
17       "facility_name": "FA0958065 - VALLEY BURGERS",
18       "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19       "violations": [
20         {
21           "id": 180,
22           "violation_code": "23",
23           "description": "Vermin or animal contamination observed",
24           "severity": "MAJOR",
25           "points_deducted": 4,
26           "observed_on": "2025-09-24"
27         }
28       ]
29     }
30   ]

```

Results table

Page size	Version	SQL stmts/req	p50 (ms)	p95 (ms)	p99 (ms)
10	naive	11	40.98	83.14	104.87
10	fixed	2	23.46	46.82	51.83
50	naive	51	109.7	134.98	145.61
50	fixed	2	17.17	29.62	34.64
200	naive	201	439.71	582.49	653.19
200	fixed	2	30.78	52.84	61.17

3.6 Raw data for all 180 requests

page_size	version	request_index	status	sql_queries	client_latency_ms	server_time_ms	records_returned
10	naive	1	200	11	59.76	33.333	10
10	fixed	1	200	2	24.893	11.323	10
10	naive	2	200	11	56.769	41.871	10
10	fixed	2	200	2	38.663	12.965	10

```
reports > hw04 > raw > nplust1_requests.csv > data
```

1	timestamp_utc,page_size,version,request_index,status,sql_queries,client_latency_ms,server_time_ms,records_returned,violations_returned,items_sha256
2	2026-09-28T08:20:15.565069+00:00,10,naive,1,200,11,59.76,33.333,10,9,c90f9b30f359d785
3	2026-09-28T08:20:15.590760+00:00,10,fixed,1,200,2,24.893,11.323,10,9,c90f9b30f359d785
4	2026-09-28T08:20:15.647864+00:00,10,naive,2,200,11,56.769,41.871,10,9,c90f9b30f359d785
5	2026-09-28T08:20:15.686826+00:00,10,fixed,2,200,2,38.663,12.965,10,9,c90f9b30f359d785
6	2026-09-28T08:20:15.732958+00:00,10,naive,3,200,11,45.73,25.175,10,9,c90f9b30f359d785
7	2026-09-28T08:20:15.752828+00:00,10,fixed,3,200,2,19.562,6.464,10,9,c90f9b30f359d785
8	2026-09-28T08:20:15.796869+00:00,10,naive,4,200,11,43.779,23.674,10,9,c90f9b30f359d785
9	2026-09-28T08:20:15.828241+00:00,10,fixed,4,200,2,30.876,13.993,10,9,c90f9b30f359d785
10	2026-09-28T08:20:15.861461+00:00,10,naive,5,200,11,32.806,20.147,10,9,c90f9b30f359d785
11	2026-09-28T08:20:15.884070+00:00,10,fixed,5,200,2,22.285,6.636,10,9,c90f9b30f359d785
12	2026-09-28T08:20:15.923652+00:00,10,naive,6,200,11,39.204,20.726,10,9,c90f9b30f359d785
13	2026-09-28T08:20:15.944020+00:00,10,fixed,6,200,2,20.114,8.421,10,9,c90f9b30f359d785
14	2026-09-28T08:20:16.001781+00:00,10,naive,7,200,11,57.477,34.514,10,9,c90f9b30f359d785
15	2026-09-28T08:20:16.029387+00:00,10,fixed,7,200,2,27.345,10.344,10,9,c90f9b30f359d785
16	2026-09-28T08:20:16.109255+00:00,10,naive,8,200,11,79.415,54.441,10,9,c90f9b30f359d785
17	2026-09-28T08:20:16.143554+00:00,10,fixed,8,200,2,33.918,10.731,10,9,c90f9b30f359d785
18	2026-09-28T08:20:16.187083+00:00,10,naive,9,200,11,43.18,27.013,10,9,c90f9b30f359d785
19	2026-09-28T08:20:16.207107+00:00,10,fixed,9,200,2,19.629,6.507,10,9,c90f9b30f359d785
20	2026-09-28T08:20:16.236254+00:00,10,naive,10,200,11,28.813,14.703,10,9,c90f9b30f359d785
21	2026-09-28T08:20:16.256386+00:00,10,fixed,10,200,2,19.876,3.753,10,9,c90f9b30f359d785
22	2026-09-28T08:20:16.289583+00:00,10,naive,11,200,11,30.262,17.379,10,9,c90f9b30f359d785
23	2026-09-28T08:20:16.342513+00:00,10,fixed,11,200,2,52.462,40.266,10,9,c90f9b30f359d785
24	2026-09-28T08:20:16.382204+00:00,10,naive,12,200,11,39.38,21.911,10,9,c90f9b30f359d785
25	2026-09-28T08:20:16.405580+00:00,10,fixed,12,200,2,22.984,9.691,10,9,c90f9b30f359d785
26	2026-09-28T08:20:16.465951+00:00,10,naive,13,200,11,60.032,30.24,10,9,c90f9b30f359d785
27	2026-09-28T08:20:16.502601+00:00,10,fixed,13,200,2,36.385,14.798,10,9,c90f9b30f359d785
28	2026-09-28T08:20:16.615550+00:00,10,naive,14,200,11,112.507,88.185,10,9,c90f9b30f359d785
29	2026-09-28T08:20:16.642375+00:00,10,fixed,14,200,2,26.508,7.715,10,9,c90f9b30f359d785
30	2026-09-28T08:20:16.687957+00:00,10,naive,15,200,11,45.157,30.391,10,9,c90f9b30f359d785
31	2026-09-28T08:20:16.711033+00:00,10,fixed,15,200,2,22.677,7.816,10,9,c90f9b30f359d785
32	2026-09-28T08:20:16.763937+00:00,10,naive,16,200,11,52.613,30.647,10,9,c90f9b30f359d785
33	2026-09-28T08:20:16.814574+00:00,10,fixed,16,200,2,50.295,10.762,10,9,c90f9b30f359d785
34	2026-09-28T08:20:16.858992+00:00,10,naive,17,200,11,43.82,26.259,10,9,c90f9b30f359d785
35	2026-09-28T08:20:16.899672+00:00,10,fixed,17,200,2,40.407,6.207,10,9,c90f9b30f359d785

3.7 Speed-up at each page size and why it grows

Page size	naive p50 (ms)	fixed p50 (ms)	Speed-up	Queries saved / request
10	40.98	23.46	1.75×	9
50	109.7	17.17	6.39×	49
200	439.71	30.78	14.29×	199

The naive implementation has the cost of making a network request to MySQL (into the Docker VM), SQL parsing and execution, and processing the result / ORM by SQLAlchemy for each related-row query, which it does $N+1$ times, resulting in a roughly linear growth of latency with the page size (≈ 2 ms per extra query: $41 \rightarrow 110 \rightarrow 440$ ms). On the other hand, the fixed implementation always has 2 statements to send; only the number of rows increases (IN list and serialized JSON), resulting in a constant latency of $\approx 17\text{--}31$ ms.

This is the reason why the speed-up ratio increases from $1.75\times$ to $6.39\times$ to $14.29\times$: at 10 rows, the overhead of each request (HTTP call, session lookup, JSON serialization) is a significant part of the time, thus the savings of 9 queries is less helpful, while at 200 rows, the 199 saved round-trips are much more important. At page size 10, the p50 latency was slightly higher than at page size 50 (23 vs 17 ms) — page size 10 run first and had a greater noise (as p95 shows); I assume that this is due to warming up / network latency in Docker.

3.8 Add one index and compare EXPLAIN before and after

```
USE s7117_rel;  
CREATE INDEX ix_violations_inspection_id ON inspection_violations (inspection_id);  
|
```

```
QUERIES = {  
    "naive_per_record_lookup": (  
        "SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on "  
        "FROM inspection_violations WHERE inspection_id = 42 ORDER BY id"  
    ),  
    "fixed_selectin_lookup_page200": (  
        "SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on "  
        "FROM inspection_violations WHERE inspection_id IN ("  
        + ", ".join(str(i) for i in range(1, 201)) + ") ORDER BY id"  
    ),  
}
```

```

=====
BEFORE - indexes on inspection_violations: [('PRIMARY', 'id')]

-- naive_per_record_lookup: EXPLAIN SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on FROM inspection_...
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table          | type | possible_keys | key      | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE      | inspection_violations | index | None          | PRIMARY | 4       | None | 200 | 10.0    | Using where |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

EXPLAIN ANALYZE:
-> Filter: (inspection_violations.inspection_id = 42) (cost=20.8 rows=20) (actual time=0.293..0.319 rows=1 loops=1)
    -> Index scan on inspection_violations using PRIMARY (cost=20.8 rows=200) (actual time=0.0205..0.29 rows=200 loops=1)

-- fixed_selectin_lookup_page200: EXPLAIN SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on FROM inspection_...
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table          | type | possible_keys | key      | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE      | inspection_violations | index | None          | PRIMARY | 4       | None | 200 | 50.0    | Using where |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

EXPLAIN ANALYZE:
-> Filter: (inspection_violations.inspection_id in (1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42))
    -> Index scan on inspection_violations using PRIMARY (cost=20.8 rows=200) (actual time=0.0204..0.104 rows=200 loops=1)

=====
AFTER - indexes on inspection_violations: [('PRIMARY', 'id'), ('ix_violations_inspection_id', 'inspection_id')]

-- naive_per_record_lookup: EXPLAIN SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on FROM inspection_...
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table          | type | possible_keys | key      | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE      | inspection_violations | ref   | ix_violations_inspection_id | ix_violations_inspection_id | 4       | const | 1 | 100.0   | None |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

EXPLAIN ANALYZE:
-> Index lookup on inspection_violations using ix_violations_inspection_id (inspection_id=42) (cost=0.35 rows=1) (actual time=0.0207..0.0217 rows=1 loops=1)

-- fixed_selectin_lookup_page200: EXPLAIN SELECT id, inspection_id, violation_code, description, severity, points_deducted, observed_on FROM inspection_...
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table          | type | possible_keys | key      | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE      | inspection_violations | index | ix_violations_inspection_id | PRIMARY | 4       | None | 200 | 100.0   | Using where |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

EXPLAIN ANALYZE:
-> Filter: (inspection_violations.inspection_id in (1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42))

```

What changed.

Prior to index existence, for the per-record lookup the WHERE predicate does not utilize any usable key, therefore MySQL will traverse the entire clustered (PRIMARY) index and filter all 200 rows (type=index, Using where; EXPLAIN ANALYZE $\approx$ 0.29 ms, 200 rows examined). After the index addition this becomes an index lookup operation (type=ref, key=ix_violations_inspection_id, ref=const, 1 row; $\approx$ 0.02 ms). For the improved version with IN (1..200) query, the index exists in the list of possible_keys, but the optimizer will stick to the plan of scanning the PRIMARY index: 162 of the total 200 rows in the table satisfy the conditions, and traversing the table sequentially is cheaper than performing 162 index scans + row lookups.

An additional re-run with the index (raw/nplus1_requests_with_index.csv) did not make much difference (the naive version without the index performed faster on the 200-row table, 635 ms, which can be considered noise): with such a small table as 200 rows, the savings from the scan ($\sim$ 0.3 ms) are insignificant compared to $\sim$ 2 ms per round-trip query, so the N+1 problem makes much more difference.

3.9 Postman: both versions at each page size



http://127.0.0.1:8817/api/nplus1/naive?page_size=10



GET



http://127.0.0.1:8817/api/nplus1/naive?page_size=10

Send

Params



Authorization

Headers (9)

Body



Pre-request Script

Tests

Settings



Accept-Encoding



gzip, deflate, br



Connection



keep-alive

Body

Cookies (1)

Headers (6)

Test Results



200 OK

35 ms

2.8 KB

Save Resp

Pretty

Raw

Preview

Visualize

JSON



```
1 {
2   "version": "naive",
3   "page_size": 10,
4   "offset": 0,
5   "sql_queries": 11,
6   "record_count": 10,
7   "violation_count": 9,
8   "items": [
9     {
10      "id": 1,
11      "facility_name": "FA0183034 - GOLDEN DINER",
12      "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13      "violations": []
14    },
15    {
16      "id": 2,
17      "facility_name": "FA0958065 - VALLEY BURGERS",
18      "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19      "violations": [
20      ]
21    }
22  ]
23 }
```



http://127.0.0.1:8817/api/nplus1/fixed?page_size=10



GET



http://127.0.0.1:8817/api/nplus1/fixed?page_size=10

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

C



Accept-Encoding



gzip, deflate, br



Connection



keep-alive

Body

Cookies (1)

Headers (6)

Test Results



200 OK

16 ms

2.8 KB

Save Resp

Pretty

Raw

Preview

Visualize

JSON



```
1  {
2    "version": "fixed",
3    "page_size": 10,
4    "offset": 0,
5    "sql_queries": 2,
6    "record_count": 10,
7    "violation_count": 9,
8    "items": [
9      {
10       "id": 1,
11       "facility_name": "FA0183034 - GOLDEN DINER",
12       "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13       "violations": []
14     },
15     {
```



http://127.0.0.1:8817/api/nplus1/naive?page_size=200



GET



http://127.0.0.1:8817/api/nplus1/naive?page_size=200

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

C

☒	Accept-Encoding	gzip, deflate, br
☒	Connection	keep-alive

Body

Cookies (1)

Headers (6)

Test Results



200 OK

304 ms

48.81 KB

Save Resp

Pretty

Raw

Preview

Visualize

JSON



```
1  {
2    "version": "naive",
3    "page_size": 200,
4    "offset": 0,
5    "sql_queries": 201,
6    "record_count": 200,
7    "violation_count": 162,
8    "items": [
9      {
10       "id": 1,
11       "facility_name": "FA0183034 - GOLDEN DINER",
12       "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13       "violations": []
14     }
15   ]
16 }
```



http://127.0.0.1:8817/api/nplus1/fixed?page_size=200



GET



http://127.0.0.1:8817/api/nplus1/fixed?page_size=200

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

C



Accept-Encoding



gzip, deflate, br



Connection



keep-alive

Body

Cookies (1)

Headers (6)

Test Results



200 OK

30 ms

48.81 KB

Save Resp

Pretty

Raw

Preview

Visualize

JSON



```
1 {
2   "version": "fixed",
3   "page_size": 200,
4   "offset": 0,
5   "sql_queries": 2,
6   "record_count": 200,
7   "violation_count": 162,
8   "items": [
9     {
10      "id": 1,
11      "facility_name": "FA0183034 - GOLDEN DINER",
12      "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13      "violations": []
14    },
15    {
16      "id": 2,
17      "facility_name": "FA0958065 - VALLEY BURGERS",
18      "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19      "violations": [
```

 **http://127.0.0.1:8817/api/nplus1/naive?page_size=50** 

GET

▼

http://127.0.0.1:8817/api/nplus1/naive?page_size=50

Send

Params ● Authorization Headers (9) Body ● Pre-request Script Tests Settings C

☒	Accept-Encoding 	gzip, deflate, br
☒	Connection 	keep-alive

Body Cookies (1) Headers (6) Test Results  200 OK 154 ms 13.05 KB [Save Resp](#)

PrettyRawPreviewVisualize

JSON ▼ 

```
1  {
2    "version": "naive",
3    "page_size": 50,
4    "offset": 0,
5    "sql_queries": 51,
6    "record_count": 50,
7    "violation_count": 45,
8    "items": [
9      {
10         "id": 1,
11         "facility_name": "FA0183034 - GOLDEN DINER",
12         "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13         "violations": []
14       },
15       {
16         "id": 2,
17         "facility_name": "FA0958065 - VALLEY BURGERS",
18         "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19         "violations": [
```

HTTP http://127.0.0.1:8817/api/nplus1/fixed?page_size=50

GET http://127.0.0.1:8817/api/nplus1/fixed?page_size=50 Send

Params • Authorization Headers (9) Body • Pre-request Script Tests Settings C

☒ Accept-Encoding ⓘ gzip, deflate, br

☒ Connection ⓘ keep-alive

Body Cookies (1) Headers (6) Test Results 200 OK 20 ms 13.05 KB Save Resp

Pretty Raw Preview Visualize JSON ↕

```

1  {
2    "version": "fixed",
3    "page_size": 50,
4    "offset": 0,
5    "sql_queries": 2,
6    "record_count": 50,
7    "violation_count": 45,
8    "items": [
9      {
10       "id": 1,
11       "facility_name": "FA0183034 - GOLDEN DINER",
12       "site_address": "8083 ALMADEN EXPY, SAN JOSE, CA 95112",
13       "violations": []
14     },
15     {
16       "id": 2,
17       "facility_name": "FA0958065 - VALLEY BURGERS",
18       "site_address": "455 SENTER RD, SAN JOSE, CA 95124",
19       "violations": [

```

Part 4 — Grounded RAG Question-Answering System

Page size	Version	SQL stmts/req	p50 (ms)	p95 (ms)	p99 (ms)
10	naive	11	40.98	83.14	104.87
10	fixed	2	23.46	46.82	51.83
50	naive	51	109.70	134.98	145.61
50	fixed	2	17.17	29.62	34.64
200	naive	201	439.71	582.49	653.19
200	fixed	2	30.78	52.84	61.17

Part 4 — Grounded RAG Question-Answering System

4.1 Corpus and index

#	Document	Source	Chunks
1	retail-food-inspection-guide.txt	LA County DPH Retail Food Inspection Guide (2025)	218
2	abc-retail-food-inspection-guide.pdf.txt	San Bernardino County ABC Retail Food Inspection Guide (2018)	78
3	print.txt	Sacramento inspection report — Pioneer House	2
4	print1.txt	Sacramento inspection report — Dat Thanh Restaurant	4
5	wikipedia-danger-zone-food-safety.txt	Wikipedia, Danger zone (food safety), CC BY-SA	3
6	wikipedia-restaurant-rating.txt	Wikipedia, Restaurant rating, CC BY-SA	3

```
# ----- corpus/index
def build_index(chunk_size: int = CHUNK_SIZE, chunk_overlap: int = CHUNK_OVERLAP):
    docs = []
    for path in CORPUS_FILES:
        docs.append(Document(text=path.read_text(encoding="utf-8"), metadata={"source": path.name}))
    splitter = SentenceSplitter(chunk_size=chunk_size, chunk_overlap=chunk_overlap)
    nodes = []
    for doc in docs:
        doc_nodes = splitter.get_nodes_from_documents([doc])
        stem = Path(doc.metadata["source"]).stem
        for i, node in enumerate(BaseNode(doc_nodes):
            node.metadata["chunk_id"] = f"{stem}#{i:03d}"
            node.id_ = node.metadata["chunk_id"]
            # Embed/LLM on the chunk text only; source travels as metadata.
            node.excluded_embed_metadata_keys = ["source", "chunk_id"]
            node.excluded_llm_metadata_keys = ["source", "chunk_id"]
        nodes.extend(doc_nodes)
    index = VectorStoreIndex(nodes, embed_model=BGEEmbedding())
    return index, docs, nodes
```

```

reports > hw04 > raw > rag_retrievals.txt
1 [2026-09-28T07:18:00.899907+00:00] HW4 RAG run; model=qwen3:8b, embed=BAAI/bge-small-en-v1.5, chunk_size=500, chunk_overlap=50, top_k=3, seed=7117
2 Corpus: 6 documents -> 308 chunks
3 abc-retail-food-inspection-guide.pdf.txt      145,155 chars   78 chunks
4 print.txt                                     3,316 chars    2 chunks
5 print1.txt                                    6,184 chars    4 chunks
6 retail-food-inspection-guide.txt             439,563 chars  218 chunks
7 wikipedia-danger-zone-food-safety.txt        4,469 chars    3 chunks
8 wikipedia-restaurant-rating.txt              6,380 chars    3 chunks
9
10 =====
11 [Q1] top-3 retrieval for: According to the Los Angeles County Retail Food Inspection Guide, what final routine inspection score range earns a food facility a "B
12 #1 score=0.8173 source=wikipedia-restaurant-rating.txt chunk_id=wikipedia-restaurant-rating#002
13 As a result of this report, in December 1997 the Board of Supervisors of Los Angeles County introduced a letter grading system already in use for some year
14 #2 score=0.8167 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#014
15 The current Letter Grade/Score Cards have been combined with the public notification requirements. A food facility will be issued one of the following by
16 #3 score=0.7840 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#009
17 Point values are totaled and subtracted from a value of 100 points. The resulting score is used in determining a facility's letter grade and reflects the
18 config C keeps ['wikipedia-restaurant-rating#002', 'retail-food-inspection-guide#014', 'abc-retail-food-inspection-guide.pdf#009']; drops []
19
20 =====
21 [Q2] top-1 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, an
22 #1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
23 == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can
24
25 =====
26 [Q2] top-3 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, an
27 #1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
28 == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can
29 #2 score=0.7699 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#196
30 Does equipment play a role in the violation? What are the ingredients in the food and how was it made? How long before the food will be served? Will the f
31 #3 score=0.7671 source=print1.txt chunk_id=print1#000
32 Pag1 Iteration Violation1b.FOOD SAFETY CERTIFICATION Observations: Food safety certificate observed to be expired. Obtain within 30 days. Code Description
33 config C keeps ['wikipedia-danger-zone-food-safety#001', 'retail-food-inspection-guide#196', 'print1#000']; drops []
34
35 =====
36 [Q2] top-5 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, an
37 #1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
38 == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can
39 #2 score=0.7699 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#196
40 Does equipment play a role in the violation? What are the ingredients in the food and how was it made? How long before the food will be served? Will the f
41 #3 score=0.7671 source=print1.txt chunk_id=print1#000
42 Pag1 Iteration Violation1b.FOOD SAFETY CERTIFICATION Observations: Food safety certificate observed to be expired. Obtain within 30 days. Code Description
43 #4 score=0.7653 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#000
44 Source: Wikipedia, "Danger zone (food safety)" (revision 1374631635), https://en.wikipedia.org/wiki/Danger\_zone\_\(food\_safety\), CC BY-SA 4.0, accessed 2026
45 #5 score=0.7650 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#022
46 Cooling shall be by one or more of the following methods: in shallow containers, separating food into smaller portions, adding ice as an ingredient, using

```

4.2 Retrieval — top-k chunks printed before the LLM call

```

def retrieve(index, question: str, k: int) -> list[dict]:
    results = index.as_retriever(similarity_top_k=k).retrieve(question)
    return [
        {
            "rank": rank,
            "score": round(float(r.score), 4),
            "source": r.node.metadata["source"],
            "chunk_id": r.node.metadata["chunk_id"],
            "text": r.node.get_content(),
        }
        for rank, r in enumerate[Any](results, start=1)
    ]

```

```

● (.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/hw04/rag.py --retrieval-only
[2026-09-28T20:36:38.784402+00:00] HW4 RAG run; model=qwen3.8b, embed=BAAI/bge-small-en-v1.5, chunk_size=500, chunk_overlap=50, top_k=3, seed=7117
Corpus: 6 documents -> 308 chunks
abc-retail-food-inspection-guide.pdf.txt      145,155 chars    78 chunks
print.txt                                    3,316 chars     2 chunks
print1.txt                                   6,184 chars     4 chunks
retail-food-inspection-guide.txt             439,563 chars   218 chunks
wikipedia-danger-zone-food-safety.txt        4,469 chars     3 chunks
wikipedia-restaurant-rating.txt              6,380 chars     3 chunks

=====
[Q1] top-3 retrieval for: According to the Los Angeles County Retail Food Inspection Guide, what final routine inspection score range earns a food facility a "B" letter grade?
#1 score=0.8173 source=wikipedia-restaurant-rating.txt chunk_id=wikipedia-restaurant-rating#002
  As a result of this report, in December 1997 the Board of Supervisors of Los Angeles County introduced a letter grading system already in use for some years in two other near
y counties, San Diego County and Riverside County. Instead of merely listing violatio...
#2 score=0.8167 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#014
  The current Letter Grade/Score Cards have been combined with the public notification requirements. A food facility will be issued one of the following by the EHS at the time
f inspection: Placard Type Description "A" Letter Grade Final routine inspection scor...
#3 score=0.7840 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#009
  Point values are totaled and subtracted from a value of 100 points. The resulting score is used in determining a facility's letter grade and reflects the status of the food f
cility at the time of the inspection. Beginning of the Inspection: 100 points Total P...
  config C keeps ['wikipedia-restaurant-rating#002', 'retail-food-inspection-guide#014', 'abc-retail-food-inspection-guide.pdf#009']; drops []

=====
[Q2] top-1 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, and according to the F
IS guidance described in the danger zone article, after how long in the danger zone should food no longer be consumed?
#1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
  == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can grow. Food safety ag
ncies, such as the United States' Food Safety and Inspection Service (FSIS), define t...

=====
[Q2] top-3 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, and according to the F
IS guidance described in the danger zone article, after how long in the danger zone should food no longer be consumed?
#1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
  == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can grow. Food safety ag
ncies, such as the United States' Food Safety and Inspection Service (FSIS), define t...

```

```

=====
[Q2] top-1 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, and according to the FS
IS guidance described in the danger zone article, after how long in the danger zone should food no longer be consumed?
#1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
  == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can grow. Food safety age
ncies, such as the United States' Food Safety and Inspection Service (FSIS), define t...

=====
[Q2] top-3 retrieval for: In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure while cooling at room temperature, and according to the FS
IS guidance described in the danger zone article, after how long in the danger zone should food no longer be consumed?
#1 score=0.7789 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
  == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can grow. Food safety age
ncies, such as the United States' Food Safety and Inspection Service (FSIS), define t...
#2 score=0.7699 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#196
  Does equipment play a role in the violation? What are the ingredients in the food and how was it made? How long before the food will be served? Will the food be saved as lefto
vers? 9. Time as a public health control; procedures & records How do employees effec...
#3 score=0.7671 source=print1.txt chunk_id=print1#000
  Popl Iteration Violations: FOOD SAFETY CERTIFICATION Observations: Food safety certificate observed to be expired. Obtain within 30 days. Code Description: Food facilities tha
t prepare, handle, or serve non-prepackaged potentially hazardous food, shall have an...
  config C keeps ['wikipedia-danger-zone-food-safety#001', 'retail-food-inspection-guide#196', 'print1#000']; drops []

```

```

=====
[Q3] top-3 retrieval for: At or below what temperature must potentially hazardous food be held when it is cold-held?
#1 score=0.8098 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#039
  [] Pasteurized milk in original, sealed containers, pasteurized milk products in original, sealed containers, raw shell eggs, and un-shucked live molluscan shellfish may be re
eived and maintained at an ambient temperature of 45°F or less. [] Hot and cold holdin...
#2 score=0.8074 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#038
  [] Food may be removed from specified holding temperatures for a cumulative time period of up to two hours to facilitate food preparation. [] Food temperature measuring devices
that are scaled only in Celsius or dually scaled in Celsius and Fahrenheit must be ac...
#3 score=0.7976 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#022
  Cooling shall be by one or more of the following methods: in shallow containers, separating food into smaller portions, adding ice as an ingredient, using an ice bath, stirrin
g frequently, using rapid cooling equipment, or using containers that facilitate heat...
  config C keeps ['retail-food-inspection-guide#039', 'retail-food-inspection-guide#038', 'abc-retail-food-inspection-guide.pdf#022']; drops []

=====
[Q3] top-5 retrieval for: At or below what temperature must potentially hazardous food be held when it is cold-held?
#1 score=0.8098 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#039
  [] Pasteurized milk in original, sealed containers, pasteurized milk products in original, sealed containers, raw shell eggs, and un-shucked live molluscan shellfish may be re
eived and maintained at an ambient temperature of 45°F or less. [] Hot and cold holdin...
#2 score=0.8074 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#038
  [] Food may be removed from specified holding temperatures for a cumulative time period of up to two hours to facilitate food preparation. [] Food temperature measuring devices
that are scaled only in Celsius or dually scaled in Celsius and Fahrenheit must be ac...
#3 score=0.7976 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#022
  Cooling shall be by one or more of the following methods: in shallow containers, separating food into smaller portions, adding ice as an ingredient, using an ice bath, stirrin
g frequently, using rapid cooling equipment, or using containers that facilitate heat...
#4 score=0.7945 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#042
  The conservative nature of the 4-hour limit for keeping foods without temperature control allows for a needed margin of safety if the temperature of the environment is higher
than 75°F. It is important to note that time/temperature control for foods held witho...
#5 score=0.7939 source=wikipedia-danger-zone-food-safety.txt chunk_id=wikipedia-danger-zone-food-safety#001
  == Description == In regards to food safety, the terms danger zone or temperature danger zone refer to temperature range in which food-borne bacteria can grow. Food safety ag
ncies, such as the United States' Food Safety and Inspection Service (FSIS), define t...

```

```

=====
[Q4] top-3 retrieval for: What happens if a restaurant gets a C?
#1 score=0.7072 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#099
  OUT - "OUT" shall be marked if a PIC is not present to perform their duties during hours of operation. Examples of violations include: • A PIC is not present during operations
l hours. [§113945(a)] • PIC allows people unnecessary to the food facility operation ...
#2 score=0.7022 source=wikipedia-restaurant-rating.txt chunk_id=wikipedia-restaurant-rating#002
  As a result of this report, in December 1997 the Board of Supervisors of Los Angeles County introduced a letter grading system already in use for some years in two other nearb
y counties, San Diego County and Riverside County. Instead of merely listing violatio...
#3 score=0.6971 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#025
  N/A - "N/A" shall be marked if a food facility only handles prepackaged food items. N/O - "N/O" shall be marked in the RARE case when there are no food employees present at th
e time of inspection. OUT - This violation category shall be marked MAJOR; example in...
  config C keeps ['retail-food-inspection-guide#099', 'wikipedia-restaurant-rating#002', 'retail-food-inspection-guide#025']; drops []

=====
[Q5] top-3 retrieval for: What score did YUMMY KITCHEN (FA0206933) at 1711 Branham Ln in San Jose receive on its most recent health inspection?
#1 score=0.6366 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#002
  Purpose This guide is designed to support EHS during retail food inspections and to help food facility operators understand essential food safety practices, principles, and pr
ocedures. Its primary aim is to ensure facilities comply with the California Retail F...
#2 score=0.6304 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#001
  The food industry and the Los Angeles County Department of Public Health's Environmental Health Division share the crucial responsibility of ensuring that food provided to con
sumers is safe and free from contaminants that could lead to foodborne illness outbre...
#3 score=0.6192 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#014
  The current Letter Grade/Score Cards have been combined with the public notification requirements. A food facility will be issued one of the following by the EHS at the time o
f inspection: Placard Type Description "A" Letter Grade Final routine inspection scor...
  config C keeps ['retail-food-inspection-guide#002', 'retail-food-inspection-guide#001', 'retail-food-inspection-guide#014']; drops []

```

```

=====
[Q6] top-3 retrieval for: What is the capital city of Australia?
#1 score=0.4568 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#183
   Los Angeles, CA 90010-2313 (213) 351-7352 Fax: (213) 351-2789 West Valley 14500 Roscoe Blvd., 5th Fl. Panorama City, CA 91401 (818) 672-2293 Fax: (818) 672-2101 West Office 61
01 W. Centinela Ave., Ste. 300, Culver City, CA 90230 (310) 410-3448 Fax: (310) 348-0...
#2 score=0.4541 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#182
   Room Panorama City, CA 91402 (818) 672-2258 Fax: (818) 672-2102 Whittier Office 12440 E. Imperial Hwy., Room 515 Norwalk, CA 90650 (562) 345-3451 Fax: (562) 409-5074 Norwalk O
ffice 12440 E. Imperial Hwy., Room 519 Norwalk, CA 90650 (562) 345-3441 Fax: (562) 40...
#3 score=0.4315 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#184
   # 145: Santa Clarita, CA 91350 (661) 287-7000 Fax: (626) 812-3017 ENVIRONMENTAL HEALTH COMMUNITY PROTECTION BRANCH OUTBREAK INVESTIGATION & EVALUATION UNIT Conducts foodb
orne illness outbreak investigations Baldwin Park 5050 Commerce Drive Baldwin Park, C...
   config C keeps []; drops [('retail-food-inspection-guide#183', 'irrelevant (score 0.457 < floor 0.6)'), ('retail-food-inspection-guide#182', 'irrelevant (score 0.454 < floor 0.6)'
), ('retail-food-inspection-guide#184', 'irrelevant (score 0.431 < floor 0.6)')]
=====

```

4.3 Three configurations: No RAG, Basic RAG, Context-engineered RAG

prompts for A (no RAG), B (raw top-3) and C (engineered)

```

def prompt_for(config: str, question: str, chunks: list[dict]) -> tuple[list, dict]:
    """Return (messages, context info) for configuration A, B or C."""
    if config == "A":
        return [HumanMessage(content=question)], {"context_chunks": [], "dropped": []}
    if config == "B":
        context = "\n\n".join(c["text"] for c in chunks)
        user = f"Use the following context to answer the question.\n\nContext:\n{context}\n\nQuestion: {question}\nAnswer:"
        return [HumanMessage(content=user)], {"context_chunks": chunks, "dropped": []}
    kept, dropped = engineer_context(chunks)
    blocks = [
        f"[Source {i}] (file: {c['source']}, chunk: {c['chunk_id']}, relevance: {c['score']:.2f})\n{c['text'].strip()}"
        for i, c in enumerate(dict(kept, start=1))
    ]
    user = "Sources:\n\n" + (" \n\n".join(blocks) if blocks else "(no relevant sources)") + f"\n\nQuestion: {question}"
    return [SystemMessage(content=GROUND_SYSTEM), HumanMessage(content=user)], {"context_chunks": kept, "dropped": dropped}

```

```

=====
[Q1] top-3 retrieval for: According to the Los Angeles County Retail Food Inspection Guide, what final routine inspection score range earns a food facility a
#1 score=0.8173 source=wikipedia-restaurant-rating.txt chunk_id=wikipedia-restaurant-rating#002
   As a result of this report, in December 1997 the Board of Supervisors of Los Angeles County introduced a letter grading system already in use for some y
#2 score=0.8167 source=retail-food-inspection-guide.txt chunk_id=retail-food-inspection-guide#014
   The current Letter Grade/Score Cards have been combined with the public notification requirements. A food facility will be issued one of the following b
#3 score=0.7840 source=abc-retail-food-inspection-guide.pdf.txt chunk_id=abc-retail-food-inspection-guide.pdf#009
   Point values are totaled and subtracted from a value of 100 points. The resulting score is used in determining a facility's letter grade and reflects th
config C keeps ['wikipedia-restaurant-rating#002', 'retail-food-inspection-guide#014', 'abc-retail-food-inspection-guide.pdf#009']; drops []
=====

```

C: drop irrelevant and duplicate chunks, order by score

```

def engineer_context(chunks: list[dict]) -> tuple[list[dict], list[dict]]:
    """Return (kept, dropped) - each dropped chunk carries a 'drop_reason'."""
    kept, dropped = [], []
    best = max((c["score"] for c in chunks), default=0.0)
    for c in sorted(chunks, key=lambda c: c["score"], reverse=True):
        if c["score"] < MIN_SCORE:
            dropped.append({**c, "drop_reason": f"irrelevant (score {c['score']:.3f} < floor {MIN_SCORE})"})
            continue
        if c["score"] < best - RELATIVE_MARGIN:
            dropped.append({**c, "drop_reason": f"irrelevant (score {c['score']:.3f} > {RELATIVE_MARGIN} below best {best:.3f})"})
            continue
        sh = _shingles(c["text"])
        dup_of = None
        for k in kept:
            ks = _shingles(k["text"])
            jac = len(sh & ks) / max(1, len(sh | ks))
            if jac >= DUP_JACCARD:
                dup_of = (k["chunk_id"], jac)
                break
        if dup_of:
            dropped.append({**c, "drop_reason": f"duplicate of {dup_of[0]} (jaccard {dup_of[1]:.2f})"})
            continue
        kept.append(c)
    return kept, dropped

```

C: grounding rules and exact refusal sentence

```

GROUNDED_SYSTEM = f"""You answer questions about restaurant food-safety inspections using ONLY the numbered sources provided.
Rules:
1. Use only facts stated in the sources. Do not use outside knowledge, even if you know the answer.
2. Cite the source number in square brackets after every fact, e.g. [Source 1] or [Source 1][Source 2].
3. If the sources do not contain enough evidence to answer, reply with exactly this sentence and nothing else: "{REFUSAL}."
4. If the question is ambiguous (for example the sources describe different counties or facilities), say that it is ambiguous and answer each case separately, v
5. Be concise: at most 5 sentences."""

```

4.4 Six-question test set and results

```

8  questions:
9  - id: Q1
10     type: answer in one chunk
11     text: >-
12         According to the Los Angeles County Retail Food Inspection Guide, what final routine inspection
13         score range earns a food facility a "B" letter grade?
14     expected_answer: A "B" letter grade is a final routine inspection score between 80 and 89.
15     expected_sources: [retail-food-inspection-guide.txt]
16     retrieval_groups:
17         - '80\s*[-|]|to\s*89'
18     answer_groups:
19         - '80\s*(?:[-|]|to|and)\s*89'
20     must_refuse: false
21
22 - id: Q2
23     type: answer needs two chunks
24     text: >-
25         In the Dat Thanh Restaurant inspection report, what temperature did the 5-gallon buckets measure
26         while cooling at room temperature, and according to the FSIS guidance described in the danger zone
27         article, after how long in the danger zone should food no longer be consumed?
28     expected_answer: >-
29         The buckets measured 155F (print1.txt, cooling methods violation); FSIS says food that remains in
30         the danger zone for more than two hours should not be consumed (Wikipedia danger zone article).
31     expected_sources: [print1.txt, wikipedia-danger-zone-food-safety.txt]
32     retrieval_groups:
33         - '155\s*°?\s°F'
34         - 'more than two hours'
35     answer_groups:
36         - '155'
37         - '(two|2)\s*hours'
38     must_refuse: false
39
40 - id: Q3
41     type: similar information across documents
42     text: >-
43         At or below what temperature must potentially hazardous food be held when it is cold-held?
44     expected_answer: >-
45         41°F or below (both county guides; the inspection reports also log 41F). The Wikipedia danger-zone
46         article gives a similar but different figure (FSIS danger zone starting at 40°F), which should not
47         replace the California 41°F rule.
48     expected_sources: [abc-retail-food-inspection-guide.pdf.txt, retail-food-inspection-guide.txt]
49     retrieval_groups:
50         - '(at or )?below 41\s*°?\s°F|41\s*°?\s°F or below'
51     answer_groups:
52         - '41'
53     must_refuse: false
54

```

```
1 {"question_id": "Q1", "question_type": "answer in one chunk", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [], "dropped": []}
2 {"question_id": "Q1", "question_type": "answer in one chunk", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.8173, "source": "document_1"}], "context_chunks": [{}]}
3 {"question_id": "Q1", "question_type": "answer in one chunk", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.8173, "source": "document_1"}], "context_chunks": [{}]}
4 {"question_id": "Q2", "question_type": "answer needs two chunks", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [{}]}
5 {"question_id": "Q2", "question_type": "answer needs two chunks", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.692, "source": "document_1"}, {"rank": 2, "score": 0.692, "source": "document_2"}], "context_chunks": [{}]}
6 {"question_id": "Q2", "question_type": "answer needs two chunks", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.692, "source": "document_1"}, {"rank": 2, "score": 0.692, "source": "document_2"}], "context_chunks": [{}]}
7 {"question_id": "Q3", "question_type": "similar information across documents", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [{}]}
8 {"question_id": "Q3", "question_type": "similar information across documents", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.7072, "source": "document_1"}, {"rank": 2, "score": 0.7072, "source": "document_2"}], "context_chunks": [{}]}
9 {"question_id": "Q3", "question_type": "similar information across documents", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.7072, "source": "document_1"}, {"rank": 2, "score": 0.7072, "source": "document_2"}], "context_chunks": [{}]}
10 {"question_id": "Q4", "question_type": "ambiguous", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [{}], "dropped": ["document_1"]}
11 {"question_id": "Q4", "question_type": "ambiguous", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.7072, "source": "document_1"}, {"rank": 2, "score": 0.7072, "source": "document_2"}], "context_chunks": [{}]}
12 {"question_id": "Q4", "question_type": "ambiguous", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.7072, "source": "document_1"}, {"rank": 2, "score": 0.7072, "source": "document_2"}], "context_chunks": [{}]}
13 {"question_id": "Q5", "question_type": "answer not in the documents", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [{}]}
14 {"question_id": "Q5", "question_type": "answer not in the documents", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.4568, "source": "document_1"}, {"rank": 2, "score": 0.4568, "source": "document_2"}], "context_chunks": [{}]}
15 {"question_id": "Q5", "question_type": "answer not in the documents", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.4568, "source": "document_1"}, {"rank": 2, "score": 0.4568, "source": "document_2"}], "context_chunks": [{}]}
16 {"question_id": "Q6", "question_type": "unrelated", "config": "A", "top_k": 3, "retrieved": [], "context_chunks": [{}], "dropped": ["document_1"]}
17 {"question_id": "Q6", "question_type": "unrelated", "config": "B", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.4568, "source": "document_1"}, {"rank": 2, "score": 0.4568, "source": "document_2"}], "context_chunks": [{}]}
18 {"question_id": "Q6", "question_type": "unrelated", "config": "C", "top_k": 3, "retrieved": [{"rank": 1, "score": 0.4568, "source": "document_1"}, {"rank": 2, "score": 0.4568, "source": "document_2"}], "context_chunks": [{}]}
19
```

```

1 hen_needed": true, "format_compliant": false, "citations": [], "ambiguity_flagged": null}
2 "context_tokens": 1310, "answer": "According to the Los Angeles County Retail Food Inspection Guide, a food facility earns a \"B\"
3 "context_tokens": 1310, "answer": "According to the Los Angeles County Retail Food Inspection Guide, a food facility earns a \"B\"
4 l above 40\u00b0F**, potentially reaching **70\u00b0F (21\u00b0C)** or higher, depending on the environment and how long it was let
5 at room temperature. According to the FSIS guidance, food should not be consumed if it remains in the danger zone for more than two
6 at room temperature [Source 3]. According to the FSIS guidance, food should not be consumed if it remains in the danger zone for mo
7
8 022"}], "dropped": [], "context_tokens": 1374, "answer": "The temperature at or below which potentially hazardous food must be held
9 022"}], "dropped": [], "context_tokens": 1374, "answer": "Potentially hazardous food must be held at or below 45\u00b0F when cold-h
10 ected.\n\n## 2. **Impact on the Restaurant:**\n- **Loss of Customers:** A \"C\" grade is often **publicly posted**, which can dete
11 restaurant gets a **C**, it means the restaurant did not meet the minimum hygiene and food safety standards set by the Los Angeles
12 restaurant gets a C, it indicates that the restaurant has a moderate number of violations and needs to improve its food safety prac
13 , "refused": true, "refused_when_needed": true, "format_compliant": false, "citations": [], "ambiguity_flagged": null}
14 1439, "answer": "The provided context does not include specific information about the health inspection score for YUMMY KITCHEN (FA
15 1439, "answer": "I cannot answer this question from the provided documents.", "llm_seconds": 119.48, "input_tokens": 1773, "output_t
16
17 he capital city of Australia is Canberra.", "llm_seconds": 89.88, "input_tokens": 1903, "output_tokens": 9, "refused_by_filter": fa
18 -inspection-guide.txt", "chunk_id": "retail-food-inspection-guide#184", "drop_reason": "irrelevant (score 0.431 < floor 0.6)"}], "d
19

```

4.5 Context-size sweep ($k = 1, 3, 5$)

reports > hw04 > [REPORT_DRAFT.md](#)

Preview
Markdown

5. k-sweep (raw/rag_k_sweep.csv):

Q	k	Relevant / irrelevant retrieved	Context tokens	B correct	C correct
Q2	1	1 / 0	387	× (missing 155°F)	× (says "not mentioned" — honest)
Q2	3	2 / 1	1311	✓	✓ with both citations
Q2	5	3 / 2	2113	✓	✓
Q3	1	0 / 1	465	× 45°F	× 45°F
Q3	3	1 / 2	1374	✓ 41°F	× 45°F
Q3	5	1 / 4	2146	× 45°F	× 45°F

reports > hw04 > raw > rag_k_sweep.csv > data

```

1 question_id,question_type,config,top_k,retrieved_chunk_ids,retrieved_scores,context_chunk_ids,dropped,context_tokens,correct_retrieval,correct_answer,grounded,refused
2 Q2,answer needs two chunks,B,1,wikipedia-danger-zone-food-safety#001,0.7789,wikipedia-danger-zone-food-safety#001,,387,False,False,True,True,False,False,,False,17.89,"
3 Q2,answer needs two chunks,C,1,wikipedia-danger-zone-food-safety#001,0.7789,wikipedia-danger-zone-food-safety#001,,387,False,False,True,True,False,True,,False,19.98,"
4 Q2,answer needs two chunks,B,3,wikipedia-danger-zone-food-safety#001;retail-food-inspection-guide#196;printl#000,0.7789;0.7699;0.7671,wikipedia-danger-zone-food-safety
5 Q2,answer needs two chunks,C,3,wikipedia-danger-zone-food-safety#001;retail-food-inspection-guide#196;printl#000,0.7789;0.7699;0.7671,wikipedia-danger-zone-food-safety
6 Q2,answer needs two chunks,B,5,wikipedia-danger-zone-food-safety#001;retail-food-inspection-guide#196;printl#000;wikipedia-danger-zone-food-safety#000;abc-retail-food-
7 Q2,answer needs two chunks,C,5,wikipedia-danger-zone-food-safety#001;retail-food-inspection-guide#196;printl#000;wikipedia-danger-zone-food-safety#000;abc-retail-food-
8 Q3,similar information across documents,B,1,retail-food-inspection-guide#039,0.8098,retail-food-inspection-guide#039,,465,False,False,True,False,True,False,,False,24.2
9 Q3,similar information across documents,C,1,retail-food-inspection-guide#039,0.8098,retail-food-inspection-guide#039,,465,False,False,True,False,True,True,,False,32.55
10 Q3,similar information across documents,B,3,retail-food-inspection-guide#039;retail-food-inspection-guide#038;abc-retail-food-inspection-guide.pdf#022,0.8098;0.8074;0.
11 Q3,similar information across documents,C,3,retail-food-inspection-guide#039;retail-food-inspection-guide#038;abc-retail-food-inspection-guide.pdf#022,0.8098;0.8074;0.
12 Q3,similar information across documents,B,5,retail-food-inspection-guide#039;retail-food-inspection-guide#038;abc-retail-food-inspection-guide.pdf#022;retail-food-insp
13 Q3,similar information across documents,C,5,retail-food-inspection-guide#039;retail-food-inspection-guide#038;abc-retail-food-inspection-guide.pdf#022;retail-food-insp
14

```

Was additional context helpful?

For question 2 (requires two chunks), k = 1 was insufficient because only the danger zone chunk was retrieved, thus neither B nor C retrieved the 155°F temperature (C accurately indicated that the sources do not mention it). At k = 3, both retrieved the information by adding printl#000 (rank 3) chunk. At k = 5, an additional 800 tokens along with two irrelevant chunks were added without any improvements. For question 3, extra context actually made the performance worse: since the leading chunk relates to the 45°F exception, the answer at k = 1 was "45°F"; k = 3 helped B to retrieve the 41°F rule; k = 5 produced four irrelevant out of five chunks, and B returned back to 45°F. Irrelevant chunks started appearing at k = 3 for both questions. The best results were observed at k = 3, hence more context does not necessarily mean better context.

4.6 Evaluation

```

def evaluate(question: dict, row: dict, context_chunks: list[dict], context_text: str) -> dict:
    """Automatic, rule-based scoring :

    correct_retrieval    every retrieval_group regex matches some chunk placed in the prompt (n/a for A and for must-refuse questions)
    refused              answer matches REFUSAL_PATTERN
    correct_answer        must_refuse -> refused; else not refused and every answer_group matches
    grounded              refused -> True; must_refuse but answered -> False; A -> n/a (no context); else every number in the answer appears in the context, and (C only)
                        every [Source n] cited exists in the prompt
    format_compliant      exact refusal sentence, or at least one valid [Source n] citation
    refused_when_needed   must_refuse -> refused; else -> not refused
    """
    answer = row["answer"]
    config = row["config"]
    refused = _matches(REFUSAL_PATTERN.pattern, answer)
    must_refuse = question["must_refuse"]

    if config == "A" or must_refuse:
        correct_retrieval = None
    else:
        correct_retrieval = all(
            any(_matches(g, c["text"]) for c in context_chunks for g in question["retrieval_groups"])
        )

    if must_refuse:
        correct_answer = refused
    else:
        correct_answer = (not refused) and all(_matches(g, answer) for g in question["answer_groups"])

    cited = [int(n) for n in re.findall(r"\\Source (\\d+)\\", answer)]
    valid_citations = bool(cited) and all(1 <= n <= len(context_chunks) for n in cited)

    if refused:
        grounded = True
    elif must_refuse:
        grounded = False
    elif config == "A":
        grounded = None
    else:
        answer_numbers = set[Any](re.findall(r"\\d+(?:\\.\\d+)?", re.sub(r"\\[Source \\d+\\]", "", answer)))
        context_numbers = set[Any](re.findall(r"\\d+(?:\\.\\d+)?", context_text))
        grounded = answer_numbers <= context_numbers and (config != "C" or valid_citations)

    exact_refusal = answer.strip().rstrip(".").strip("'").lower() == REFUSAL.lower()
    format_compliant = exact_refusal or valid_citations

    ambiguity_flagged = None
    if question.get("ambiguity_markers"):

```

Configuration	Retrieval	Accuracy	Faithfulness	Format compliance	Robustness
A - No RAG	n/a	2/6 (33%)	2/3 (67%)	0/6 (0%)	1/3 (33%)
B - Basic RAG	3/4 (75%)	4/6 (67%)	5/6 (83%)	0/6 (0%)	1/3 (33%)
C - Context-engineered RAG	3/4 (75%)	4/6 (67%)	6/6 (100%)	6/6 (100%)	2/3 (67%)

```

reports > hw04 > raw > rag_evaluation_table.csv > data
1 question_id,question_type,config,correct_retrieval,correct_answer,grounded,refused_when_needed,format_compliant,ambiguity_flagged
2 Q1,answer in one chunk,A,,False,,True,False,
3 Q1,answer in one chunk,B,True,True,True,True,False,
4 Q1,answer in one chunk,C,True,True,True,True,True,
5 Q2,answer needs two chunks,A,,False,True,False,False,
6 Q2,answer needs two chunks,B,True,True,True,True,False,
7 Q2,answer needs two chunks,C,True,True,True,True,True,
8 Q3,similar information across documents,A,,True,,True,False,
9 Q3,similar information across documents,B,True,True,True,True,False,
10 Q3,similar information across documents,C,True,False,True,True,True,
11 Q4,ambiguous,A,,False,,True,False,True
12 Q4,ambiguous,B,False,False,True,True,False,False
13 Q4,ambiguous,C,False,False,True,True,True,False
14 Q5,answer not in the documents,A,,True,True,True,False,
15 Q5,answer not in the documents,B,,True,True,True,False,
16 Q5,answer not in the documents,C,,True,True,True,True,
17 Q6,unrelated,A,,False,False,False,False,
18 Q6,unrelated,B,,False,False,False,False,
19 Q6,unrelated,C,,True,True,True,True,
20 |

```

4.7 Analysis

The retrieval retrieved correct chunks for questions 1, 2 and 3, but did not for question 4. In case of Q1, the grade table for LA from the retail-food-inspection-guide#014 took second position. The Wikipedia history paragraph about LA letter grades was also on the same level (0.817 and 0.817) as grade table from above, although both B and C gave answer to "80-89". Two documents were required for Q2: Dat Thanh report and the danger-zone article. Both appeared at $k \geq 3$, since print1#000 took the third position. At $k = 1$ both configurations did not have information about the 155°F. B declared that "the report does not provide" the information about the measurement, while it can be found in the document. C stated that provided sources did not contain the information, but nevertheless, the chunk was answered by it. Q4 ("What happens if a restaurant gets a C?") was not solved by retrieval. Letter "C" matched checklist chunks of violation *Categories* and not grading section, so no configuration could answer the question.

The contextual modifications that helped were the grounding rules, the refusal sentence, and source labels. Source labeling made all answers to question C citable (format compliance: 100% compared to 0% for B). Grounding rules transformed Q5 into a refusal and ensured the model would not answer Q6 using memory. The score floor filtered out all chunks of Q6 prior to generation (all scores ≤ 0.46), but none of those for on-topic questions. There were no significant differences between bge scores of relevant and distractor chunks – both had scores in the 0.01-0.03 range. Therefore, the score threshold is a crude off-topic filter, but not a relevance ranker. De-duplication didn't happen: an overlap of 50 tokens in 500-token chunks is well below the Jaccard 0.5 criterion. Smaller context helped indirectly by reducing token count. C used 17% less tokens on average (1,122 vs 1,356) since Q6 context was ignored, and was faster. Larger k didn't help: at $k = 5$, Q3 had four off-target chunks, and B again gave 45°F.

AI use

1. What I used an AI assistant for, and what I did myself

I used Claude Code (Anthropic) as a coding agent in VS Code. It:

- used AI to write the reports and debugging errors
- to draft sql migration queries
- wrote the seed, benchmark, EXPLAIN, metrics and the Makefile
- I was not able to install MySql directly using homebrew on my machine so took help from AI to resolve the issue and use it using Docker container

What I did myself: Wrote the code files and reviewed each and every code generated through AI. Choose the questions and verified their answers. Created the database migrations and ran all the apis in Postman. Integrated apis to frontend and tested all the functionalities from React UI.

2. One AI-produced output that was wrong, and one thing I verified independently

Wrong: the first UpdateRecord.jsx loaded the record inside useEffect with no cleanup.

In development, React StrictMode runs effects twice. The second, late response overwrote the name

already typed into the form, so **Update** sent the old name. The page still redirected to Home with no error.

Verified independently: the RAG evaluator's grounded flag. Config C answered Q3 with "45°F".

It was marked grounded, because 45 appears in the cited chunk, but the answer was wrong. I read

chunk retail-food-inspection-guide#039 myself. It is about the 45°F exception for shellfish, eggs and milk. The 41°F cold-holding rule was in Source 3 (abc-retail-food-inspection-guide.pdf#022).

3. How I detected the problem / verified the result

- Update bug: headless-Chromium end-to-end test (log in → create → edit → delete) failed when waiting

for the renamed row to appear; API log revealed PUT /api/records/5003 → 200, meaning backend accepted

the request; body contained original value. Two GET /api/records/5003 requests per page load were in the

API log too; the double effect.

- Grounded vs. correct: I checked the grounded and correct_answer columns in raw/rag_evaluation_table.csv and identified rows which were grounded but incorrect. Then I opened the chunk text for each row in raw/rag_retrievals.txt.
- I additionally verified the N+1 numbers in the code rather than in the table. In verify.py there is an assertion of exactly N+1 statements (naive) versus 2 (fixed), with equal items. EXPLAIN proves the index is used correctly.

4. What I changed and why it works now

- Update/Delete pages: the fetch effect now sets an ignore flag locally in cleanup, and discards any response that arrives after cleanup is done. Only the most recent fetch can populate the form. The

inputs are disabled until then. The same test case in Chromium passes: create → rename → delete.

- Evaluation: I did not change the automatic scores, but report the Q3 case as a known limitation. The grounded evaluation ensures that the facts in the answer actually exist in the context. This does

not ensure whether the model has selected the right fact. Thus, faithfulness (100% for C) and accuracy (4/6)

are to be considered together.

- Other suggestions I evaluated and kept:

MySQL works in Docker, since Homebrew does not include bottles anymore for Intel macOS and fails in building from source in the test stage;

bge-small-en-v1.5 is used instead of MiniLM of HW3, as it uses 512 token windows, which fit the 500 token chunk size without truncating;

CORS allows origins to be listed explicitly, as a * origin cannot be combined with credentialed (cookie) requests.